PV101 Day 7 - Battery Storage vs. Solar Needs and Wire Sizing

Student Worksheet - print ready

Name: _____ Team: _____ Date: _____ Instructor: _____

Today's big idea

A solar system has two different needs: making energy and storing energy. A panel is the faucet. A battery is the tank. Wire is the road. Fuses are the guardrails. If one part is undersized, the whole system becomes unreliable.

Core formulas for today

Watts = Volts x Amps
Watt-hours per day = Watts x Hours used per day
Battery watt-hours = Battery volts x Amp-hours
Usable storage = Battery watt-hours x usable percent
Solar per day = Panel watts x peak sun hours x system efficiency
Amps = Watts / Volts
Fuse rule: protect the wire first. Fuse size must not exceed what the wire can safely carry.

Part 1 - Load list: How much energy do we need?

Directions: Use Watts x Hours = Watt-hours. Fill in the last column. Round to the nearest whole watt-hour.

Load / device	Volts	Amps or Watts given	Hours per day	Wh/day
LED lights	12 V	2 A	4 hr	_____
Roof vent fan	12 V	3 A	3 hr	_____
Phone charging station	12 V	24 W	2 hr	_____
Water pump	12 V	5 A	0.5 hr	_____
Small inverter load	120 V AC	150 W	1 hr	_____
Your added load: _____	____	____	____	_____

Total daily energy need: _____ Wh/day

One sentence summary: This system needs about _____ Wh each day because _____.

Part 2 - Battery storage: How big is the tank?

Given battery for this exercise

12.8 V LiFePO4 battery, 50 Ah. For class design work, use 80% usable storage unless the instructor says otherwise.

1. Battery watt-hours = 12.8 V x 50 Ah	_____ Wh
2. Usable storage = battery Wh x 0.80	_____ Wh usable
3. If your daily load is the Part 1 total, will one battery cover one day?	Yes / No because _____
4. How many of these batteries would give roughly two days of usable storage?	_____ batteries
5. Why is it not smart to drain a battery to empty every day?	_____

Quick check: More battery storage helps at night and on cloudy days. More panel helps recharge during daylight. They solve different problems.

Part 3 - Solar needs: How fast can we refill the tank?

Class assumption

Peak sun hours are not the same as sunrise-to-sunset hours. Today, use 4 peak sun hours and 75% system efficiency. That 75% includes real-world losses from heat, wiring, controller behavior, angle, and imperfect conditions.

Formula: Solar Wh/day = Panel watts x peak sun hours x efficiency

Panel setup	Formula	Solar Wh/day	Enough for your Part 1 load?
One 100 W panel	100 W x 4 hr x 0.75	_____	Yes / No
One 200 W panel	200 W x 4 hr x 0.75	_____	Yes / No
Two 200 W panels	400 W x 4 hr x 0.75	_____	Yes / No
Your design: _____ W	____ W x 4 hr x 0.75	_____	Yes / No

Explain the mismatch if there is one:

_____

If the solar makes less energy than the loads use each day, what happens over several days?

_____

Part 4 - Wire sizing: How big does the road need to be?

Important safety note

This is a training worksheet, not a final code design. Real installations require manufacturer instructions, fuse ratings, temperature ratings, wire type, length, voltage drop, and applicable electrical code. Today we are practicing the thinking process.

Step 1: Find current. Use Amps = Watts / Volts when watts are given.

Circuit	Voltage	Load	Current	First wire-size thought
LED lights	12 V	2 A	2 A	Small load, still fuse it
Vent fan	12 V	3 A	3 A	Small load, length matters
Water pump	12 V	5 A	5 A	Higher startup surge
200 W panel into controller	About 12 V system	200 W	_____ A	Controller/wire must handle current
300 W inverter on 12 V battery	12 V	300 W	_____ A	Much higher DC current than people expect
600 W inverter on 12 V battery	12 V	600 W	_____ A	Large DC current, short thick wire

What surprised you about inverter current on the 12 V side?

_____

Part 5 - Simplified wire and fuse thinking

Rule of thumb for this class

Wire must be sized for current and distance. Fuse protects the wire. A fuse that is too large can allow the wire to overheat before the fuse opens. A fuse that is too small may nuisance-blow during normal use.

Training current range	Training wire-size thought	Fuse thinking
0-5 A	Small branch circuit; keep runs neat and protected	Fuse above normal load, below wire limit
5-15 A	Medium branch circuit; check length and voltage drop	Common small PV/load range
15-30 A	Larger DC circuit; wire size matters fast	Controller/inverter input territory
30 A+	High current; short runs, thick wire, serious protection	Do not guess
Unknown	Stop and calculate or look up rating	No mystery fuses

Part 6 - Team design challenge

Scenario: You are planning a small camper/RV learning system. It should run the listed loads for one normal day and recover with solar during decent sun. Use the assumptions from this worksheet.

1. Total daily load from Part 1	_____ Wh/day
2. Battery usable storage per 12.8 V 50 Ah battery	_____ Wh usable
3. Number of batteries you recommend	_____
4. Panel watts you recommend	_____ W
5. Solar Wh/day using 4 sun hours and 75% efficiency	_____ Wh/day
6. Highest-current circuit in your design	_____
7. First circuit you would fuse and why	_____
8. One thing you would verify with a meter before turning it on	_____

Part 7 - Reflection and exit ticket

1. In your own words, what is the difference between solar need and battery/storage need?

_____
_____

_____
_____

2. Why can a small 12 V inverter pull a surprisingly large amount of current from a battery?

_____
_____

_____
_____

3. What is one mistake a beginner might make when choosing wire or fuse size?

_____
_____

_____
_____

4. Today's professional habit: Before replacing parts, I will _____.

_____
_____

_____
_____

Exit phrase

Panels make energy. Batteries store energy. Wires carry current. Fuses protect wires. Professionals verify before they build.